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RESEARCH

Atmospheric water vapour distribution and moisture transport over Europe and their impact on the humidity conditions over Poland

Ewelina Krawczyk[1]

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Abstract

The amount of atmospheric water vapour and its transport were analysed to rate the intensity and main directions of moisture fluxes incoming to Poland. The total column water vapour (TCWV) was used to show the total amount of water vapour in the entire atmosphere, whereas the specific humidity (SH) allow to show vertical variation of the moisture properties. Also, the SH fluxes were applied to verify the main directions of water vapour transport at the pressure level of 925 hPa, 850 hPa and 700 hPa over Europe. The Atlantic Ocean is the relevant source of water vapour, which is advected to Europe due to the western circulation dominant in mid-latitudes. Huge supplies of water vapour and higher wind speeds affect the strongest fluxes there. Thus, the strongest SH fluxes come from the west (the monthly mean in July is up to 61 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$ in Lodz). It is also the most frequent direction of atmospheric water vapour supplies over Poland. Simultaneously, long-term trends in TCWV and SH are mostly insignificant. There is a slight increase in the amount of water vapour in the subpolar region in the colder half of the year, and over Central, Eastern and Northern Europe in summer. These results show that the increasing saturation deficit and more frequent dry events are not caused by the change in the amount of water vapour. This provides the ground for claiming that the drying of the atmosphere is affected mainly by increasing temperature.

1 Introduction

Water vapour is an important component of the atmosphere. Its amount is differentiated spatially and dependent on the altitude (Palarz and Celiński-Mysław 2020). In general, the water vapour amount is decreasing as the altitude increases. Its amount is mostly higher in warmer regions due to stronger evaporation and an increasing capacity as the air temperature increases (Kunkel et al. 2013). It also means that the amount of atmospheric water vapour is strongly related to air temperature (Jackson et al. 2006; Scholz and Lora 2024); more water vapour is observed in the warmer air horizontally and vertically. According to Wypych and Bochenek (2018), the relevance of water vapour in the atmosphere is based on its properties of thermal absorption as a greenhouse gas. The atmospheric moisture also affects the heat balance through the latent heat exchange during the change of physical state (Ruprecht 1996; Sherwood et al. 2010).

The air temperature is gradually increasing (Lee et al. 2021), which has a relevant impact on the moisture content. According to the Clausius-Clapeyron equation, the water vapour capacity of the atmospheric air is dependent on the air temperature. With a 1 °C increase in air temperature, the water vapour storage capacity for the atmosphere grows about 7% (Trenberth 2011). Therefore, the conditions needed to achieve a saturation state will change with increasing temperature. Moreover, the rise of air temperature favours the evaporation to strengthen (Findell et al. 2019). As a result, the amount of water vapour described by the absolute or specific humidity is increasing (Dessler and Davis 2010; Stoy et al. 2022). Mostly, the increase in the amount of water vapour is relatively lower than its maximum capacity, so the saturation deficit increases (Cheng et al. 2023; Wen et al. 2024; Clark et al. 2026; Krawczyk 2026). Therefore, the moisture properties in the near-surface atmosphere are definitely changing as a result of the increase in air temperature. On the other hand, another source of the water vapour is the advection from other regions (Seager et al. 2010). So the observations of water vapour changes and

✉ Ewelina Krawczyk
ewelina.krawczyk@edu.uni.lodz.pl

[1] Doctoral School of Exact and Natural Sciences, University of Lodz, Lodz, Poland

its transport in the whole troposphere are relevant for the detailed assessment of moisture content.

The research on atmospheric moisture properties over Europe shows significant relations with conditions over the Atlantic Ocean and the Arctic. Wypych and Bochenek (2018) point to the impact of atmospheric circulation on moisture properties over Europe, especially in winter. It determines the direction of water vapour advection and causes anomalies in the amount of atmospheric water in the higher troposphere related to the properties of incoming air masses. They show the domination of western water vapour transport from the Atlantic Ocean to Europe in the entire vertical profile of the troposphere. A part of water vapour from the Atlantic Ocean is also transported by the atmospheric rivers to Western Europe (Ramos et al. 2016; Zavadoff and Kirtman 2020). According to Zavadoff and Kirtman (2020), higher temperatures and the intensification of evaporation might lead to the strengthening of atmospheric rivers by additional water vapour resources. The location of Europe defines the main properties for incoming air masses. The western advection to Europe is more humid than the eastern advection of continental air masses (Phillips and McGregor 2001). Besides, the relation in vapour transport between Europe and the Arctic is considered in detail. The evaporation is quite low in the Arctic; most of the water vapour comes by advection from lower latitudes (Jakobson and Vihma 2010; Nygård et al. 2019). Hocke and Wang (2023) stated that the northern advection brings low supplies of atmospheric water.

The previous research described contrary trends, which might be an effect of the different long-term periods used. The amount of water vapour in the atmospheric air column (precipitable water) is mostly increasing in Poland, according to Wibig and Siedlecki (2007). Ojrzyńska et al. (2022) stated that the significant trends are mostly observed in lower layers of the troposphere in spring (negative trends) and summer (positive trends). Whereas, Degirmendžić and Kożuchowski (2017) point to the decreasing trend of the average precipitable water.

The research described in this paper is focused on the amount of atmospheric water vapour and its transport over Europe. The key aim is to show the main sources of atmospheric water vapour advected to Poland and their relations with general moisture transport over Europe. This analysis improves the research by a wider data range and a different approach to the described issue. Also, the research will allow for improving recent research on the humidity conditions in Poland (Krawczyk 2026) by the analysis of moisture sources. Moreover, the rarely applied methods and the observations of relations between the amount of water vapour and evaporation improve the recent research on moisture content over Europe. These observations are based on the differences between chosen pressure levels of the troposphere in changes of moisture content and horizontal transport of water vapour, instead of the integrated water vapour in columns within pressure levels applied in many analyses. The trends in water vapour might allow to detect dependencies with climate change and detect whether the drought episodes are affected by the deficit of water vapour in the atmosphere or are an effect of increasing air temperature. The spatial variation is analysed in different altitudes to compare the range of changes and define their reason.

2 Data and methods

The longitudinal range of the research area is between 30°W and 45°E, and the latitudinal range is from 25°N to 75°N (Fig. 1). Therefore, the area considered in this research includes a slightly wider spatial range than Europe. The reason for choosing this area range and the simultaneous aim of this research is to show the impact on atmospheric water vapour amount and sources of its advection to Poland. Therefore, the impact of marine air from the Atlantic Ocean and continental air from Western Asia was considered. Also, more probable directions of water vapour fluxes are included (especially a pretty huge part of the Atlantic Ocean).

The monthly data from ERA5 reanalysis were used to verify spatial distribution and trends in atmospheric water vapour properties in 1966–2020. The datasets at a resolution of 0.25° x 0.25° were prepared by the European Centre for Medium-Range Weather Forecasts (ECMWF) and came from the Copernicus Climate Change Service (C3S) (Hersbach et al. 2020). For the total column water vapour and actual evaporation analysis the single level database was used, whereas the data from three pressure levels (925, 850 and 700 hPa) were used for specific humidity amount and horizontal fluxes analysis (Copernicus Climate Change Service 2023a; 2023b).

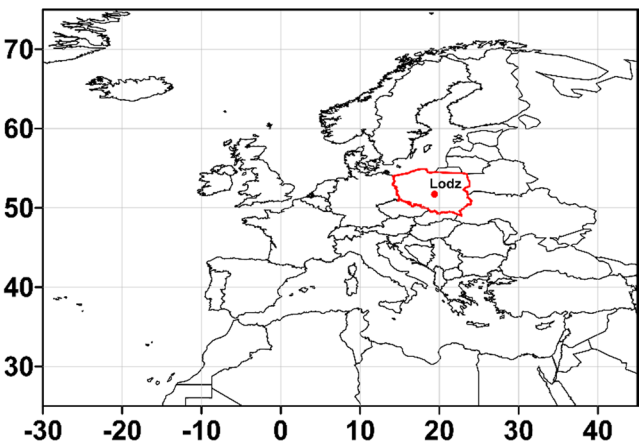


Fig. 1 Map of the research area range

The described results are based on the analysis of total column water vapour (*TCWV*; also known as precipitable water) and specific humidity (*SH*) spatial distribution, focusing on the multi-year monthly mean values and trends, which were calculated using Climate Data Operators (Schulzweida 2023). Additionally, the impact of evaporation on the moisture content has been described, comparing the spatial and seasonal differentiation of the monthly mean of daily evaporation to the TCWV conditions. Long-term trends were stated on the basis of linear model coefficients using the "stats" R package (R Core Team 2022). The statistical significance of trends was verified using the p-value (with the confidence level = 95%). Using the specific humidity and zonal and meridional wind components, the horizontal specific humidity fluxes (*SHFLUX*) were calculated according to the formula:

$$SHFLUX = \sqrt{(q \bullet u)^2 + (q \bullet v)^2}$$

where q – specific humidity [g·kg^{-1}], u – zonal wind component [m·s^{-1}], v – meridional wind component [m·s^{-1}] (Wypych and Bochenek 2018). Mean directions of specific humidity fluxes were calculated on the basis of long-term averages of wind components. Factors determining the humidity conditions in Poland were described using mean specific humidity fluxes from each of 8 directions and frequency of advection from each direction to the geographical coordinates of Lodz (Central Poland) (51°43'24''N, 19°23'59''E; marked with a red dot in Fig. 1).

Results were visualised on the maps prepared in the Surfer 10 program using the Ordinary Point Kriging interpolation method. The properties of water vapour advection to Central Poland were shown using polar plots for the mean specific humidity fluxes and the frequency of directions from which water vapour came.

2.1 Results and analysis

2.1.1 Amount of water vapour in the atmosphere over Europe

Total column water vapour as a parameter of air humidity allows for showing spatial changes in the vertically summarised amount of water vapour in the atmosphere. The dominant source of water vapour over Europe is the Atlantic Ocean (Gimeno et al. 2010; Wypych et al. 2018). For most of the year, the TCWV is much higher over the ocean, especially in lower latitudes, where it can reach up to 32 kg/m^2 (Fig. 2). Additionally, the second important source of atmospheric water vapour is the Mediterranean Sea basin (Wypych et al. 2018). However, its impacts strengthen seasonally from May to October, when the TCWV might be about 30 kg/m^2. Over the land, the amount of atmospheric

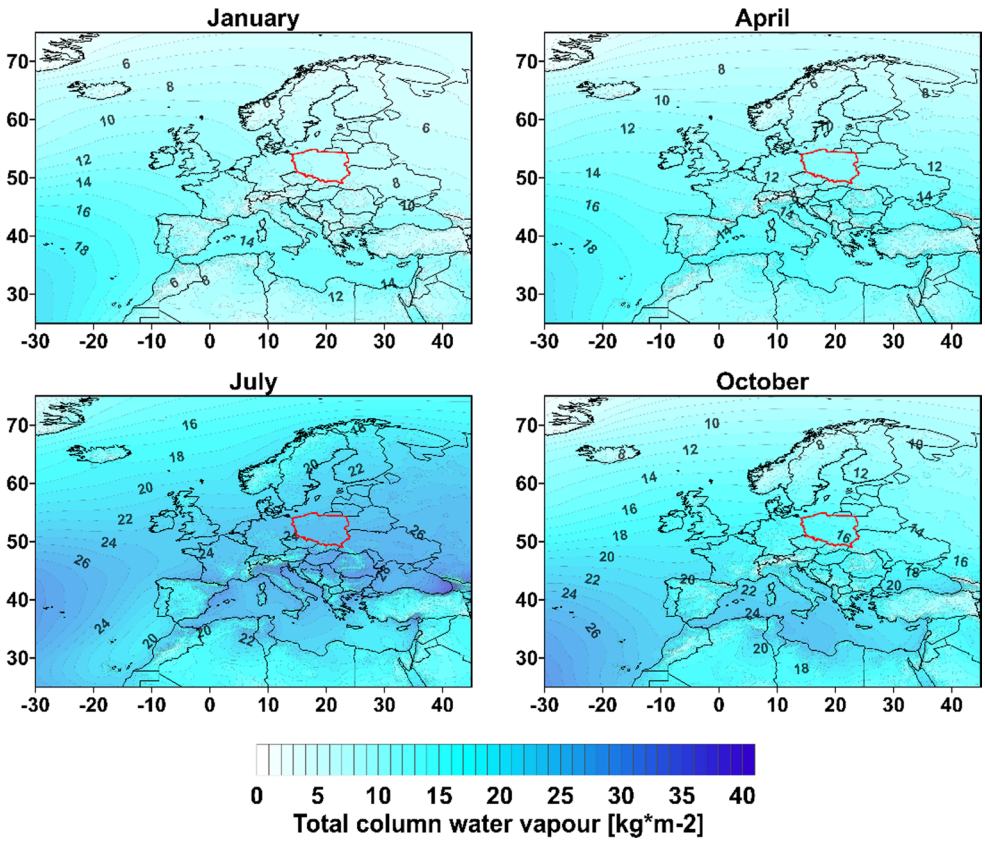


Fig. 2 Monthly mean of total column water vapour over Europe in January, April, July and October in 1966–2020

water vapour is lower and decreases in highlands and mountainous regions. Simultaneously, the TCWV varies annually there. In Poland, the impact of inland conditions and lower temperature causes the decrease in TCWV to mostly 8–10 kg/m^2 in winter months. Whereas in summer months, the TCWV increases to 24–26 kg/m^2, which is an effect of evapotranspiration strengthening and water vapour transport from other regions of Europe. The TCWV is permanently lower in the subarctic region.

Evaporation changes seasonally and impacts the main source of water vapour over Europe, which is especially pronounced by the differences in inland evaporation (Fig. 3). In January and October, inland evaporation mostly does not exceed 2 mm/day of water equivalent, and the disparities between inland and oceanic evaporation intensity are notable (mostly 3–6 mm/day over the water bodies). The differences between inland and oceanic evaporation are gradually reduced in April; lower evaporation occurs in Northern Europe, whereas its intensity in Western and Central Europe is similar to conditions over the Atlantic Ocean and the Mediterranean Sea (2–3 mm/day of water equivalent). In July, stronger evaporation over the land areas than over the ocean is pronounced (2–4 mm/day, locally higher), however the disparities with conditions over water bodies are large due to relatively lower evaporation over the Atlantic Ocean. Nonetheless, the relationship between TCWV and evaporation is particularly important due to the changes over inland Europe. Also, the moderate correlation for these dependencies was detected (Pearson's correlation from 0.51 in April to 0.68 in July). These observations point to the substantial impact of local evaporation on the moisture content over Europe, whereas the water vapour transport from the Atlantic Ocean seems to play the key role in moisture conditions in winter.

The amount of water vapour decreases towards higher altitudes, and the spatial distribution of atmospheric water vapour becomes less varying in the higher altitudes than near the surface (Fig. 4). It is caused by more intensive land-atmosphere exchange of water vapour in the boundary layer, mostly by the evaporation processes. At the pressure level of 925 hPa, the specific humidity reaches up to 14 g/kg in summer months and up to 8 g/kg in winter months over the Atlantic Ocean in lower latitudes. Over continental Europe, the amount of water vapour decreases gradually in the northeastern direction. However, the specific humidity changes more slowly and is quite high over the land in the lower atmosphere because of the additional source of the vapour from stronger evapotranspiration in the summer months. In Poland, the specific humidity changes from slightly below 3 g/kg in January and February to 8 g/kg in July and August. At the pressure levels of 850 hPa and 700 hPa, the differences in the specific humidity over the land and ocean become smaller, which is related to weaker land-atmosphere interactions. Over the Atlantic Ocean, the

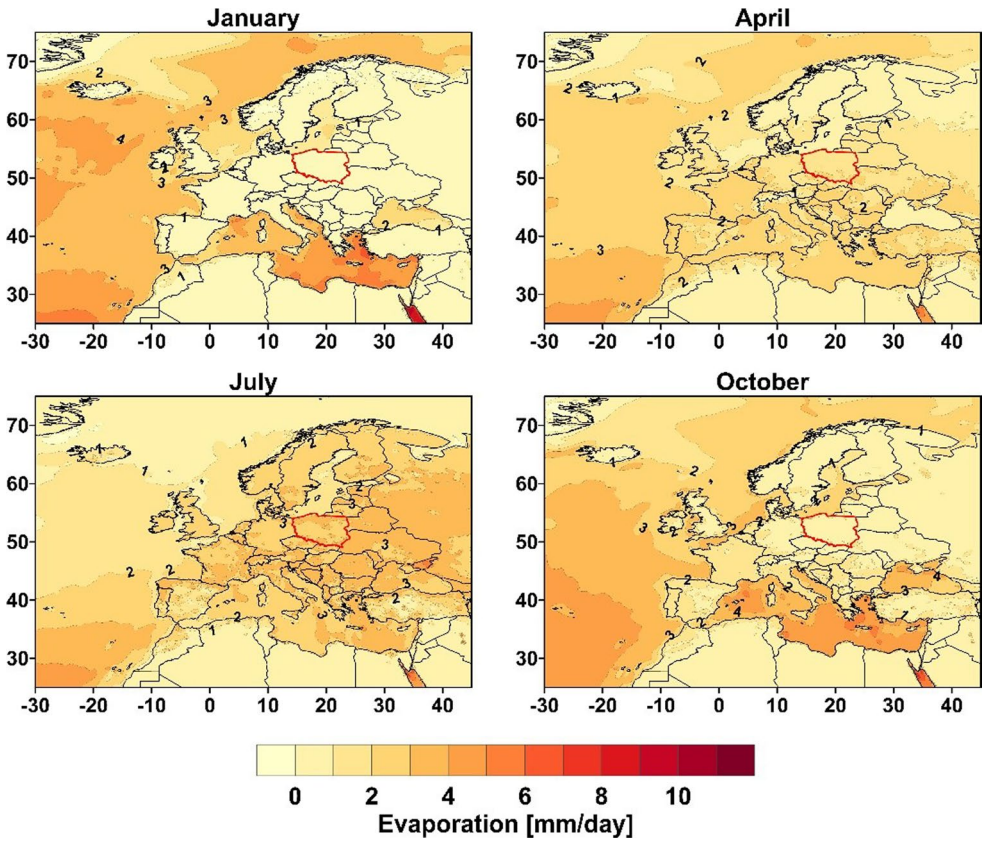


Fig. 3 Monthly mean of actual evaporation (mm/day) over Europe in January, April, July and October in 1966–2020

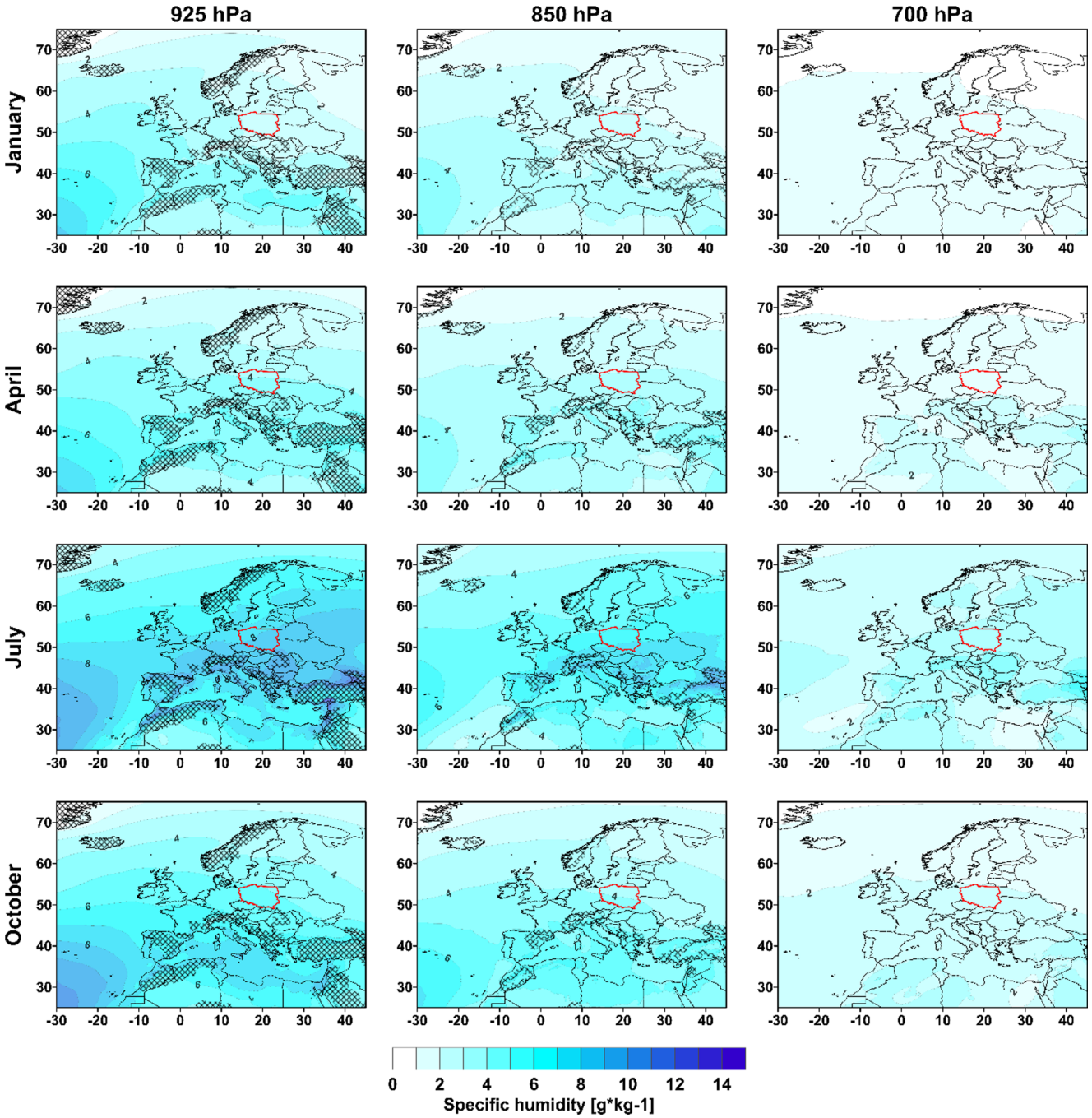


Fig. 4 Distribution of the monthly mean of specific humidity over Europe in January, April, July and October at the pressure levels of 925, 850 and 700 hPa; the shaded areas were excluded due to the lowest altitude of the pressure levels below the surface level

specific humidity reaches up to 8 g/kg in summer and up to 5 g/kg in winter at the 850 hPa pressure level. Whereas at the pressure level of 700 hPa, the specific humidity is mostly below 4 hPa throughout the year over Europe. Therefore, the annual variation of specific humidity is smaller at higher pressure levels. In Poland, the specific humidity changes annually from 2 to 7 g/kg at the pressure level of 850 hPa and 1–4 g/kg at the 700 hPa pressure level.

3 Water vapour transport over Europe

The main factors of water vapour transport character are circulation patterns and the distribution of large water reservoirs (the Atlantic Ocean in this case). Mid-latitude zone circulation is mostly determined by the western winds. It is visible at all pressure levels, especially at higher layers of the atmosphere, where the friction effect does not impact

the modifications of wind direction. The most intensive SH fluxes were observed over the Atlantic Ocean at all pressure levels (Fig. 5). However, the strongest fluxes are in lower layers of the atmosphere due to the higher amount of water vapour there. In summer months, they are mostly up to 60 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$, locally reaching 80 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$ in lower latitudes. Simultaneously, the friction causes a decrease in wind speed and direction changes; this effect is more prominent over land, where SH fluxes are below 30 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$ in most areas. Unlike the specific humidity distribution, the lowest SH fluxes are in spring. At the pressure level of 850 hPa, the SH fluxes reach 50 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$ over the ocean

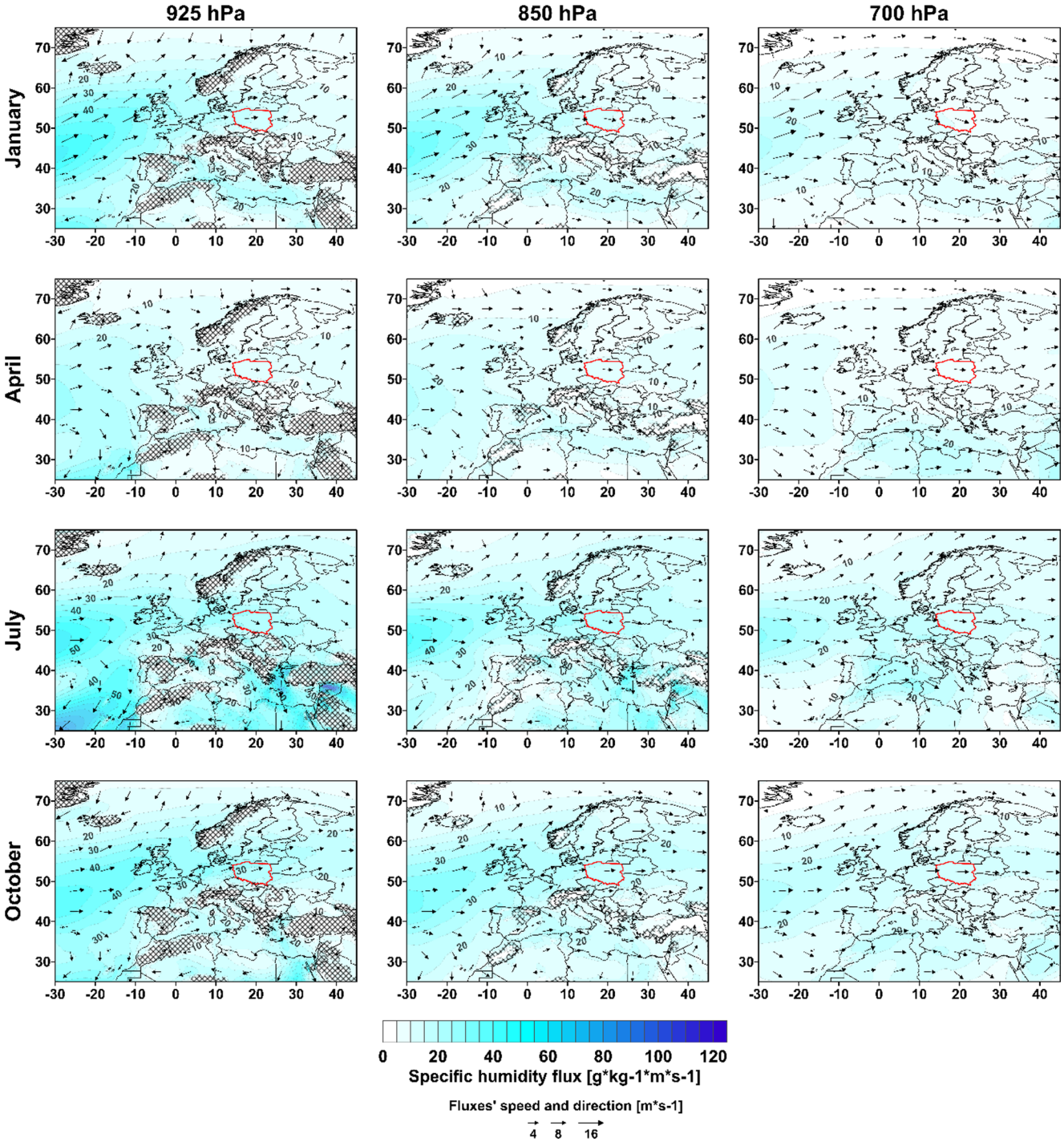


Fig. 5 Monthly mean of specific humidity fluxes in January, April, July and October and characteristics of horizontal advection of water vapour over Europe at the pressure levels of 925, 850 and 700 hPa; the shaded areas were excluded due to the lowest altitude of the pressure levels below the surface level

and rarely exceed 25 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$ over the land. Locally over the Mediterranean Sea, fluxes might be slightly higher in summer and reach 40 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$. Similar to conditions at the pressure level of 925 hPa, the inland water vapour transport is lower in spring than in the other seasons and does not exceed 20 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$. At the pressure level of 700 hPa, the wind speed is higher than at the lower atmosphere and the western direction is more prominent. Despite stronger wind, the SH fluxes are lower due to the amount of water vapour decreasing with altitude. Over the Atlantic Ocean, fluxes rarely exceed 30 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$ and are mostly below 20 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$ over the land. Annual variation in the intensity of SH flux is less visible at 700 hPa pressure level than in the near-surface layers of the atmosphere. The water vapour advection is slightly more intensive in summer months, while very low fluxes are observed in the subarctic region in the colder half of the year.

Horizontal transport to Poland is dominated by the western directions (W, NW or SW) in the atmospheric air (Fig. 6). In all months, the summarised frequency of advection from western, northwestern and southwestern directions is not lower than 49% at the pressure level of 925 hPa. At higher altitudes, it increases to >65% (850 hPa) and >72% (700 hPa). During the year, the weakest western advection is observed in May at all of the analysed pressure levels, while in the colder months it is more frequent. Eastern advection is quite rare; its observation is more probable in spring. Annually, SH fluxes are lower from January to April, regardless of the altitude. Whereas, strengthening of the water vapour transport takes place from July to November. At the pressure level of 925 hPa, stronger vapour fluxes come from western directions and often exceed 20 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$. Additional sources of water vapour in July and August come from southern and southeastern advection; these SH fluxes are quite prominent and reach up to 30 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$. At the pressure level of 850 hPa, the SH fluxes are slightly lower than near the Earth's surface. However, the water vapour transport is still quite intensive from June to October. In comparison with the SH fluxes at the pressure level of 925 hPa, more water vapour is advected from the West, and other transport sources are less efficient. Western fluxes can exceed 20 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$ from June to December, reaching up to 35 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$ in July. Meanwhile, the SH fluxes are much weaker at the pressure level of 700 hPa, hardly ever reaching more than 20 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$ even in summer. From December to May, the fluxes are up to 12 $g \cdot kg^{-1} \cdot m \cdot s^{-1}$. The western advection of water vapour is more prominent in comparison to other directions, sometimes being the primary source of atmospheric vapour, like from November to January.

4 Long-term changes in the amount of atmospheric water vapour

In the long-term perspective, significant changes in TCWV occur seasonally and regionally in limited areas. Mostly, the trends are increasing, but these changes are rarely strong (Fig. 7). The increase in TCWV is observed in the subarctic region from September to February, usually up to 0.5 kg/m^2/10 years. In most months of the year, increasing trends are also visible over the Atlantic Ocean. There, higher trends over larger areas are observed in the warmer half of the year, often exceeding 0.4 kg/m^2/10 years. The slight increase in TCWV (up to 0.4 kg/m^2/10 years) in Southern and Central Europe is developing gradually from April to August. Decreasing trends in TCWV are observed seasonally, usually from November to April in the Mediterranean region and near the Black Sea (up to -0.5 kg/m^2/10 years).

Long-term trends in SH at the pressure levels also show changes in the subarctic region in the colder half of the year (Fig. 8). However, the area with statistically significant changes and the magnitude of trends shrink in higher altitudes. At the pressure level of 925 hPa, the SH increases slightly, up to 0.3 g/kg/10 years, but these changes don't exceed 0.2 g/kg/10 years in most cases. These trends also include Northern Europe. At the pressure levels of 850 and 700 hPa, trends of SH in the subarctic region are low, rarely exceeding 0.1 g/kg/10 years. The decreasing SH is observed in the Mediterranean Sea basin in winter and summer months at the pressure levels of 925 and 850 hPa. However, these changes are not stronger than 0.3 g/kg/10 years, and they are isolated to small regions. These changes are not as visible at the pressure level of 700 hPa. In the warmer half of the year, SH trends at the pressure levels of 925 and 850 hPa are similar to TCWV trends. Additional changes are observed in the Iberian Peninsula and in the Black Sea basin. The decrease in SH at previously mentioned pressure levels is more often observed than a change in the amount of water vapour for the total column of the atmosphere. The summer trends in SH over Central and Western Europe at the pressure level of 850 hPa are widely expanded in comparison to the pressure level of 925 hPa, but they are weaker than in the near-surface atmosphere. At the pressure levels of 850 and 700 hPa, the SH does not change as intensively as at the pressure level of 925 hPa. The trends are rarely stronger than 0.1 g/kg/10 years (both increasing and decreasing) at the pressure level of 700 hPa. They are mostly concentrated over the Atlantic Ocean and the subarctic region, and also in different parts of the Mediterranean Sea basin seasonally.

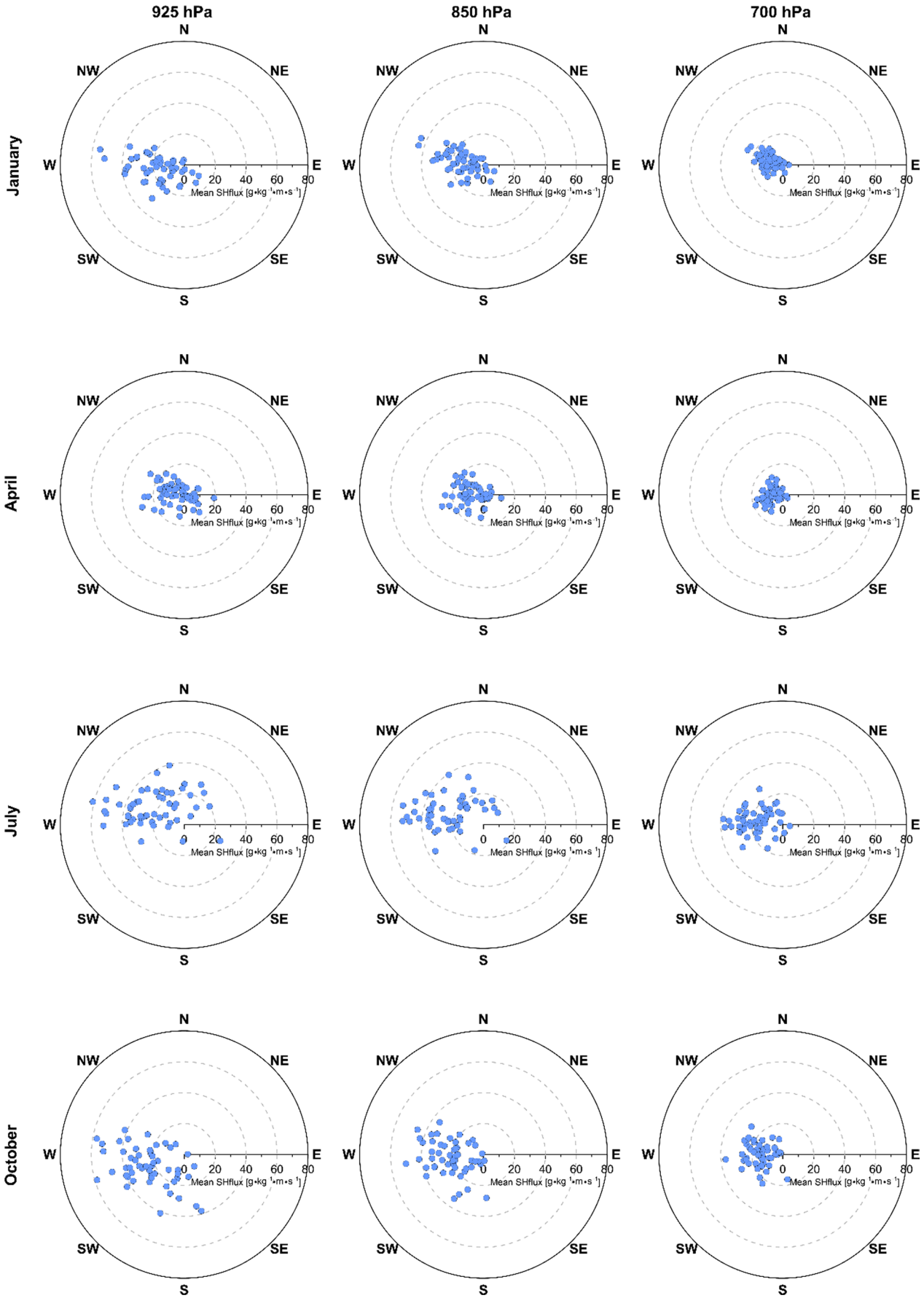

925 hPa
850 hPa
700 hPa
January
April
July
October
N
NE
E
SE
S
SW
W
NW
0
20
40
60
80
Mean SHflux [g•kg⁻¹•m•s⁻¹]

Fig. 6 Monthly mean fluxes of specific humidity (SH flux; $g \cdot kg^{-1} \cdot m \cdot s^{-1}$) in January, April, July and October from each of 8 directions in 1966–2020 over Lodz (Central Poland) at the pressure levels of 925, 850 and 700 hPa

5 Discussion

In general, the highest total column water vapour and specific humidity are observed over the Atlantic Ocean, especially in lower latitudes (TCWV > 30 kg/m^2 and SHUM > 10 g/kg). Spatial variation is similar for the total column water vapour and specific humidity at each pressure level and in all seasons. More water vapour is observed over the ocean and decreases gradually in the northern and northeastern directions (Figs. 2 and 4). Therefore, the prominent supplies of water vapour for inland Europe are transported by advection from the Atlantic Ocean. Trenberth et al. (2011) stated that the hydrologic cycle is based on the intensive evaporation over the oceans and precipitation in the inland regions. Because of the stronger evaporation over water reservoirs (Fig. 3), the amount of atmospheric water vapour is larger over the Atlantic Ocean than over Europe (Fig. 2). According to Held and Soden (2006), most of the water vapour in the atmospheric column is concentrated in the lower parts of the atmosphere. It is related to the vertical profile of air temperature and its impact on saturation water vapour pressure, and also the intensive water vapour exchange between Earth's surface and the atmosphere. Therefore, the specific humidity is the highest at the level of 925 hPa compared with levels of 850 hPa and 700 hPa (Fig. 4). The amount of water vapour gradually decreases in the higher altitudes, which also impacts lower transport of the atmospheric water despite the stronger wind speed.

The most intensive specific humidity fluxes are observed over the Northern Atlantic Ocean, which is determined not only by the amount of water vapour but also by stronger advection at these latitudes (Fig. 5). The lower horizontal wind speed impacts weaker fluxes near the tropic and subpolar latitudes. Additionally, weaker fluxes are over inland Europe as a result of lower water vapour supplies and lower wind weakened by the friction. Seager et al. (2010) point to mechanical and thermodynamic factors affecting the water vapour transport. From a mechanical point of view, most of Europe is exposed to the western circulation, which is dominant in midlatitudes due to global circulation mechanisms. Therefore, the western water vapour fluxes are more frequent, especially above the boundary layer where the friction is negligible (Fig. 6). On the other hand, the thermodynamic conditions are related to evaporation processes, which affect the atmospheric water vapour accumulation. The type of Earth's surface and its land cover impact the atmospheric water vapour resources by the water availability and its potential evaporation capability. Nygård et al.

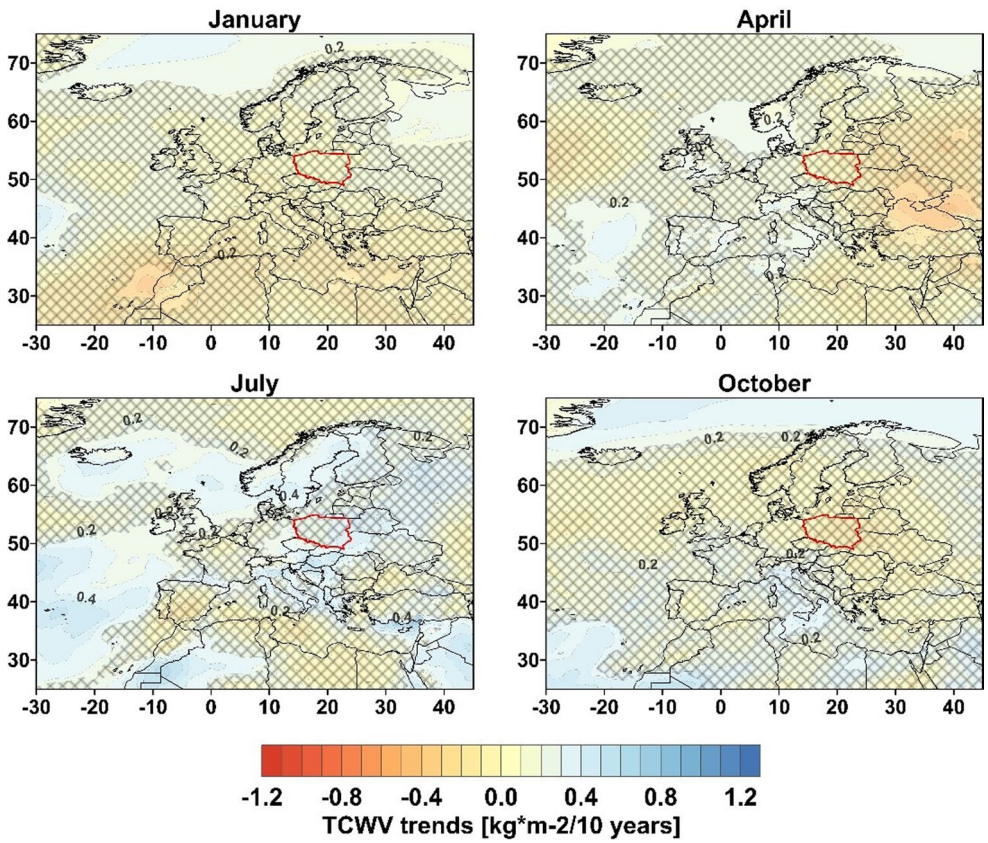


Fig. 7 Trends of monthly total column water vapour over Europe in January, April, July and October in 1966–2020; shaded areas are statistically insignificant

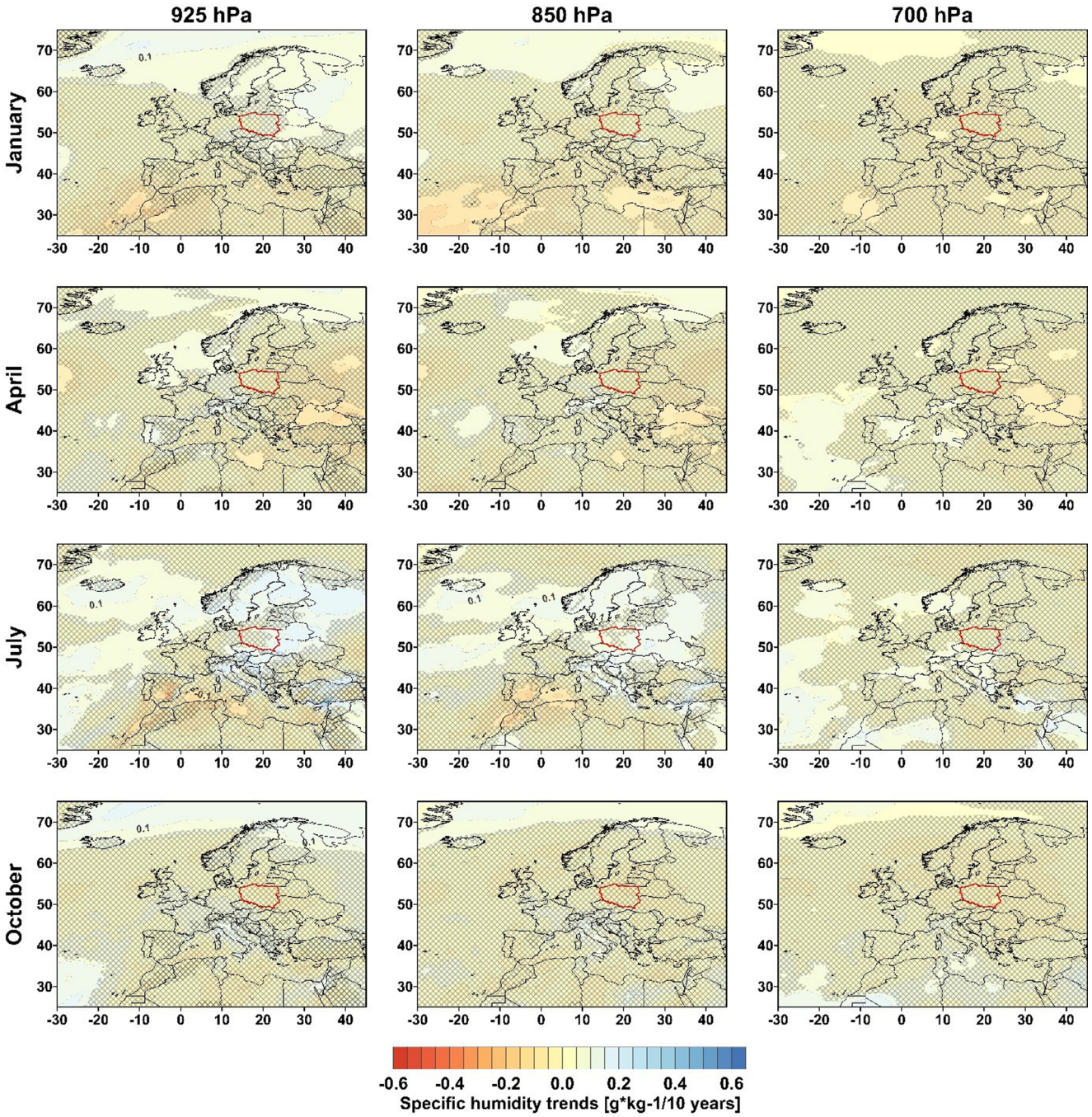


Fig. 8 Trends of monthly specific humidity in January, April, July and October at the pressure levels of 925, 850 and 700 hPa over Europe in 1966–2020; shaded areas are statistically insignificant

(2019) also pay attention to the exchange of atmospheric water vapour between Europe and the Arctic by the southward transport. On the other hand, Hocke and Wang (2023) stated that the water vapour transport from the Arctic is quite low, which is in agreement with lower northern and northeastern fluxes. In the European background, the specific humidity fluxes from the Arctic are much weaker (Fig. 5). The vertical variation of vapour transport is related to the amount of water vapour at pressure levels. The specific humidity fluxes are stronger in the lower troposphere and decrease at higher pressure levels despite increasing wind speed. Therefore, water vapour resources have a crucial impact on the intensity of atmospheric water transport in general. Also, Held and Soden (2006) stated that larger reservoirs of water vapour in the near-surface atmosphere lead to more intense advection at lower heights. Higher specific

humidity fluxes over the Atlantic Ocean point to the larger supplies of atmospheric water vapour evaporated from the oceanic surface layer. Additionally, lower friction over the ocean contributes to the increase in wind speed, which also enlarges the moisture transport.

Long-term trend in specific humidity and TCWV over the research area are mostly insignificant. However, there are a few visible dependencies in the seasonality of significant trends, which are slightly increasing in most cases. In autumn and winter, the increasing trend is relevant in the subarctic region and locally over the Atlantic Ocean. Whereas in summer, the increase in specific humidity and TCWV is noticed over the Central, Eastern and Northern Europe (Figs. 7 and 8). Humidity properties are dependent on the thermal conditions. Higher air temperature favours evaporation strengthening, and so the precipitable water is larger in warmer seasons (Fig. 2) (Rinke et al. 2019). Also, the increasing air temperature in the long-term perspective are in relation with positive trends in evaporation strengthening and growth in the amount of water vapour (Fig. 7). In the multiyear perspective, these changes lead to modifications in the entire hydrological cycle (Seager et al. 2010). The positive trends in precipitable water (Fig. 7) are mostly related to stronger evaporation in the warmer atmosphere (Zavadoff and Kirtman 2020; Tabassum et al. 2024). However, it is worth noticing that the evaporated water vapour is delivered mostly to lower parts of the atmosphere, so the amount of specific humidity varies vertically (Fig. 4). Held and Soden (2006) highlighted that the stronger evaporation in warmer climates causes an increase in the amount of water vapour mostly in the lower atmosphere. In agreement with them, the increase in specific humidity at the pressure level of 925 hPa is relevant in summer over central, eastern and northern regions of Europe, which in a large part correlates with an increase in precipitable water (Figs. 7 and 8). The processes over the Atlantic Ocean are particular; they play a crucial role in water vapour transport to Europe. In the long-term perspective, the increase in precipitable water and specific humidity is observed there. Serreze et al. (2012) also mentioned that this trend is visible in the northern parts of the Atlantic Ocean, which might support the advection of water vapour to the Arctic.

The moisture advection in Poland is dominated by the western directions. Their frequency increases at higher altitudes, where the impact of friction and surface properties is much lower than in the near-surface layer of the atmosphere. Specific humidity fluxes come to Poland also from the southern (in April and October) and northern (in July) regions (Fig. 6). Poland's location causes the simultaneous impact of different factors on the climate. Western advection is dominant in midlatitudes and affects the moisture transport from the Atlantic Ocean. From this direction, the intensive specific humidity fluxes come to Central Poland (Fig. 6). Therefore, each change in humidity exchange in Western Europe might impact moisture properties in Poland. The trends in the amount of water vapour in the atmospheric column are slightly increasing, but mostly insignificant in most cases, which is in agreement with Wibig and Siedlecki (2007). The exception is the southeastern region of Poland, where the precipitable water significantly increases in summer (Fig. 7). According to Araszkiewicz et al. (2021), it might be an effect of stronger convective processes in the continental climate of Southeastern Poland during summer, which boost the evaporation.

6 Conclusions

The atmospheric water vapour participates in thermodynamic and circulation processes, so the moisture issues are extensive and complex. The water vapour transport is an important factor regulating not only the air humidity conditions, but also the entire hydrological cycle. In general, the moisture conditions over Europe have been mostly dependent on the humidity fluxes incoming from the Atlantic Ocean. Large atmospheric water supplies in oceanic air masses and western circulation in mid-latitudes are favourable to water vapour advection to Europe. The general increasing trend in the total column water vapour is supposed to be an effect of increasing air temperature and stronger evaporation. It is contributed by changes in specific humidity in the near-surface atmosphere, whereas the amount of atmospheric water at higher altitudes is less changeable. Simultaneously, the significant trends in the amount of water vapour are mostly observed over large reservoirs. Only in summer, the increase in specific humidity or precipitable water is significant over the inland areas of Central, Eastern and Northern Europe. Therefore, the changes in the amount of atmospheric water vapour or its transport are mostly insignificant and do not have a large impact on the hydrological cycle.

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Data availability The datasets analysed during the current study are available in the Copernicus Climate Change Service (C3S) Climate Data Store (CDS) repository, https://cds.climate.copernicus.eu/datasets.

Declarations

Competing interests The authors declare no competing interests.

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References

Araszkiewicz A, Kiliszek D, Mierzwiak M, Da Costa N, Szołucha JM (2021) GPS-based multi-temporal variation in precipitable water over the territory of Poland. Remote Sens. 2021, 13, 2960. https://doi.org/10.3390/rs13152960

Cheng M, Zuo Z, Lin Z et al (2023) The decadal abrupt change in the global land vapor pressure deficit. Sci China Earth Sci 66:1521–1534. https://doi.org/10.1007/s11430-022-1117-x

Clark B, Visioni D, Kothari S, Lerdau M (2026) Land models likely underestimate the impact of future atmospheric dryness on European tree growth. Geophys Res Lett 53:e2026GL122759. https://doi.org/10.1029/2026GL122759

Copernicus Climate Change Service (2023a) ERA5 monthly averaged data on pressure levels from 1940 to present. Copernicus Clim Change Service (C3S) Clim Data Store (CDS). https://doi.org/10.24381/cds.6860a573. Accessed on 11-06-2025

Copernicus Climate Change Service (2023b) ERA5 monthly averaged data on single levels from 1940 to present. Copernicus Clim Change Service (C3S) Clim Data Store (CDS). https://doi.org/10.24381/cds.f17050d7. Accessed on 22-10-2025

Degirmendžić J, Kożuchowski K (2017) Mediterranean cyclones, the atmospheric moisture content and precipitation in Poland. Geographia Polonica 2017, Vol. 90, I. 1. Institute of geography and spatial organization Polish academy of sciences pp. 5–20. https://doi.org/10.7163/GPol.0075

Dessler AE, Davis SM (2010) Trends in tropospheric humidity from reanalysis systems. J Geophys Res 115:D19127. https://doi.org/10.1029/2010JD014192

Findell KL, Keys PW, van der Ent RJ, Lintner BR, Berg A, Krasting JP (2019) Rising Temperatures Increase Importance of Oceanic Evaporation as a Source for Continental Precipitation. J Clim 32(22):7713–7726. https://doi.org/10.1175/JCLI-D-19-0145.1

Gimeno L, Drumond A, Nieto R, Trigo RM, Stohl A (2010) On the origin of continental precipitation. Geophys Res Lett 37:L13804. https://doi.org/10.1029/2010GL043712

Held IM, Soden BJ (2006) Robust Responses of the Hydrological Cycle to Global Warming. J Clim 19(21):5686–5699. https://doi.org/10.1175/JCLI3990.1

Hersbach H, Bell B, Berrisford P, Hirahara S, Horányi A, Muñoz-Sabater J, Nicolas J, Peubey C, Radu R, Schepers D, Simmons A, Soci C, Abdalla S, Abellan X, Balsamo G, Bechtold P, Biavati G, Bidlot J, Bonavita M, De Chiara G, Dahlgren P, Dee D, Diamantakis M, Dragani R, Flemming J, Forbes R, Fuentes M, Geer A, Haimberger L, Healy S, Hogan RJ, Hólm E, Janisková M, Keeley S, Laloyaux P, Lopez P, Lupu C, Radnoti G, de Rosnay P, Rozum I, Vamborg F, Villaume S, Thépaut JN (2020) The ERA5 global reanalysis. Q J R Meteorol Soc 146: 1999–2049. https://doi.org/10.1002/qj.3803

Hocke K, Wang W (2023) Observation of an Extremely Dry Atmospheric Air Column above Bern. Climate 11(3):63. https://doi.org/10.3390/cli11030063

Jackson DL, Wick GA, Bates JJ (2006) Near-surface retrieval of air temperature and specific humidity using multisensor microwave satellite observations. J Geophys Res 111:D10306. https://doi.org/10.1029/2005JD006431

Jakobson E, Vihma T (2010) Atmospheric moisture budget in the Arctic based on the ERA-40 reanalysis. Int J Climatol 30:2175–2194. https://doi.org/10.1002/joc.2039

Krawczyk E (2026) The impact of climate warming on the air humidity characteristics and their long-term perspective in Poland. Geographia Polonica 99(2):167–187. https://doi.org/10.7163/GPol.0322

Kunkel KE, Karl TR, Easterling DR, Redmond K, Young J, Yin X, Hennon P (2013) Probable maximum precipitation and climate change. Geophys Res Lett 40:1402–1408. https://doi.org/10.1002/grl.50334

Lee JY, Marotzke J, Govindasamy B, Cao L, Corti S, Dunne JP, Engelbrecht F, Fischer E, Fyfe JC, Jones C, Maycoch A, Muterni J, Ndiaye O, Panickal S, Zhou T (2021) Future global climate: scenario-based projections and near-term information [w:] V Masson-Delmotte. In: Zhai P, Pirani A, Connors SL, Péan C, Berger S, Caud N, Chen Y, Goldfarb L, Gomis MI, Huang M, Leitzell K, Lonno E, Matthews JBR, Maycock TK, Waterfield T, Yelekçi O, Yu R, Zhou B (red.) (eds) Climate Change 2021: The Physical Science Basis. Contribution of working group I to the sixth assessment report of the intergovernmental panel on climate change. Cambridge University Press, pp 553–672

Nygård T, Graversen RG, Uotila P, Naakka T, Vihma T (2019) Strong Dependence of Wintertime Arctic Moisture and Cloud Distributions on Atmospheric Large-Scale Circulation. J Clim 32(24):8771–8790. https://doi.org/10.1175/JCLI-D-19-0242.1

Ojrzyńska H, Błaś M, Kryza M (2022) Precipitable water content over Poland. Atmosphere 13:988. https://doi.org/10.3390/atmos13060988

Palarz A, Celiński-Mysław D (2020) Low-tropospheric humidity inversions over Europe: spatiotemporal variability and relations to temperature inversions' occurrence. Theor Appl Climatol 141:967–978. https://doi.org/10.1007/s00704-020-03250-z

Phillips ID, McGregor GR (2001) Western European Water Vapor Flux–Southwest England Rainfall Associations. J Hydrometeorol 2(5):505–524. https://doi.org/10.1175/1525-7541(2001)002%3C0505:WEWVFS%3E2.0.CO;2

R Core Team (2022) R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/

Ramos AM, Tomé R, Trigo RM, Liberato MLR, Pinto JG (2016) Projected changes in atmospheric rivers affecting Europe in CMIP5 models. Geophys Res Lett 43:9315–9323. https://doi.org/10.1002/2016GL070634

Rinke A, Segger B, Crewell S, Maturilli M, Naakka T, Nygård T, Vihma T, Alshawaf F, Dick G, Wickert J, Keller J (2019) Trends of Vertically Integrated Water Vapor over the Arctic during 1979–2016: Consistent Moistening All Over? J Clim 32(18):6097–6116. https://doi.org/10.1175/JCLI-D-19-0092.1

Ruprecht E (1996) Atmospheric water vapour and cloud water: an overview. Adv Space Res 18(7):5–16. https://doi.org/10.1016/0273-1177(95)00282-0

Scholz SR, Lora JM (2024) Atmospheric rivers cause warm winters and extreme heat events. Nature 636:640–646. https://doi.org/10.1038/s41586-024-08238-7

Schulzweida U (2023) CDO User Guide (2.3.0). Zenodo. https://doi.org/10.5281/zenodo.10020800

Seager R, Naik N, Vecchi GA (2010) Thermodynamic and Dynamic Mechanisms for Large-Scale Changes in the Hydrological Cycle in Response to Global Warming. J Clim 23(17):4651–4668. https://doi.org/10.1175/2010JCLI3655.1

Serreze MC, Barrett AP, Stroeve J (2012) Recent changes in tropospheric water vapor over the Arctic as assessed from radiosondes and atmospheric reanalyses. J Geophys Res 117:D10104. https://doi.org/10.1029/2011JD017421

Sherwood SC, Roca R, Weckwerth TM, Andronova NG (2010) Tropospheric water vapor, convection, and climate. Rev Geophys 48:RG2001. https://doi.org/10.1029/2009RG000301

Stoy PC, Roh J, Bromley GT (2022) It's the heat and the humidity: The complementary roles of temperature and specific humidity to recent changes in the energy content of the near-surface atmosphere. Geophys Res Lett 49:e2021GL096628. https://doi.org/10.1029/2021GL096628

Tabassum A, Park K, Shin J, Jin HG, Baik JJ (2024) Long-term changes in temperature, specific humidity, and precipitation in Bangladesh revealed by ERA5 data. Theoret Appl Climatol 155(3):1915–1925. https://doi.org/10.1007/s00704-023-04732-6

Trenberth KE (2011) Changes in precipitation with climate change. Climate Res 47(1–2):123–138

Trenberth KE, Fasullo JT, Mackaro J (2011) Atmospheric Moisture Transports from Ocean to Land and Global Energy Flows in Reanalyses. J Clim 24(18):4907–4924. https://doi.org/10.1175/2011JCLI4171.1

Wen R, Qin M, Jiang P, Yang F, Liu B, Zhu M, Fang Y, Tian Y, Shang B (2024) Vegetation and evapotranspiration responses to increased atmospheric vapor pressure deficit across the global forest. Atmosphere 15:408. https://doi.org/10.3390/atmos15040408

Wibig J, Siedlecki M (2007) Przestrzenny i czasowy rozkład zawartości wody opadowej w atmosferze nad Europą (1958–2005). In Wahania Klimatu w Różnych Skalach Przestrzennych i Czasowych; Piotrowicz, K., Twardosz, R., Eds.; Instytut Geografii i Gospodarki Przestrzennej, Uniwersytet Jagielloński: Kraków, Poland, pp. 195–204. (In Polish)

Wypych A, Bochenek B (2018) Vertical structure of moisture content over Europe. Adv Meteorology 13:3940503. https://doi.org/10.1155/2018/3940503

Wypych A, Bochenek B, Różycki M (2018) Atmospheric Moisture Content over Europe and the Northern Atlantic. Atmosphere 9(1):18. https://doi.org/10.3390/atmos9010018

Zavadoff BL, Kirtman BP (2020) Dynamic and Thermodynamic Modulators of European Atmospheric Rivers. J Clim 33(10):4167–4185. https://doi.org/10.1175/JCLI-D-19-0601.1

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